

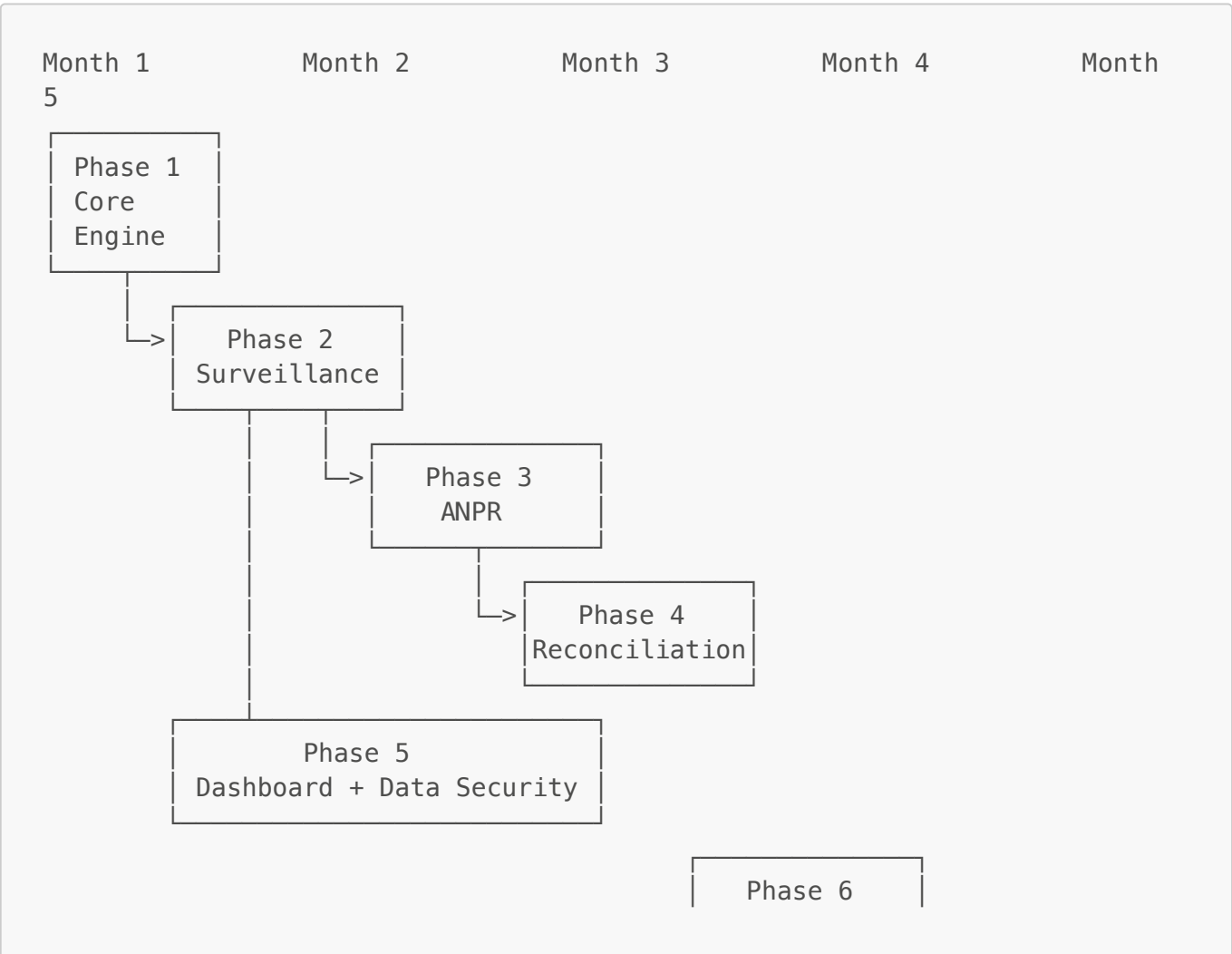
# Q-Vision: Detailed Implementation Plan

## All Phases — Technical & Operational Breakdown

**Version:** 1.0  
**Date:** March 2026  
**Project:** Quarry Material Monitoring & Anti-Theft System

### Overview

| Phase   | Name                                           | Duration  | Depends On         |
|---------|------------------------------------------------|-----------|--------------------|
| Phase 1 | Core Intelligence Engine                       | Month 1   | —                  |
| Phase 2 | Live Surveillance & Auto-Capture               | Month 1–2 | Phase 1            |
| Phase 3 | Vehicle Identification (ANPR)                  | Month 2–3 | Phase 2            |
| Phase 4 | Material Reconciliation & Theft Detection      | Month 3–4 | Phase 2, Phase 3   |
| Phase 5 | Owner Dashboard, Mobile Access & Data Security | Month 3–4 | Phase 1 (parallel) |
| Phase 6 | Deployment, Training & Support                 | Month 4–5 | All phases         |



## Phase 1: Core Intelligence Engine

**Duration:** Month 1  
**Investment:** ₹12,00,000  
**Status:** POC completed (12mm, 20mm classification demo working)

### What This Phase Delivers

The brain of the system — the computer vision engine that looks at stone aggregate and determines what grade it is, whether it's contaminated, and how confident the system is about the result.

### Detailed Tasks

#### 1.1 Stone Grade Classification

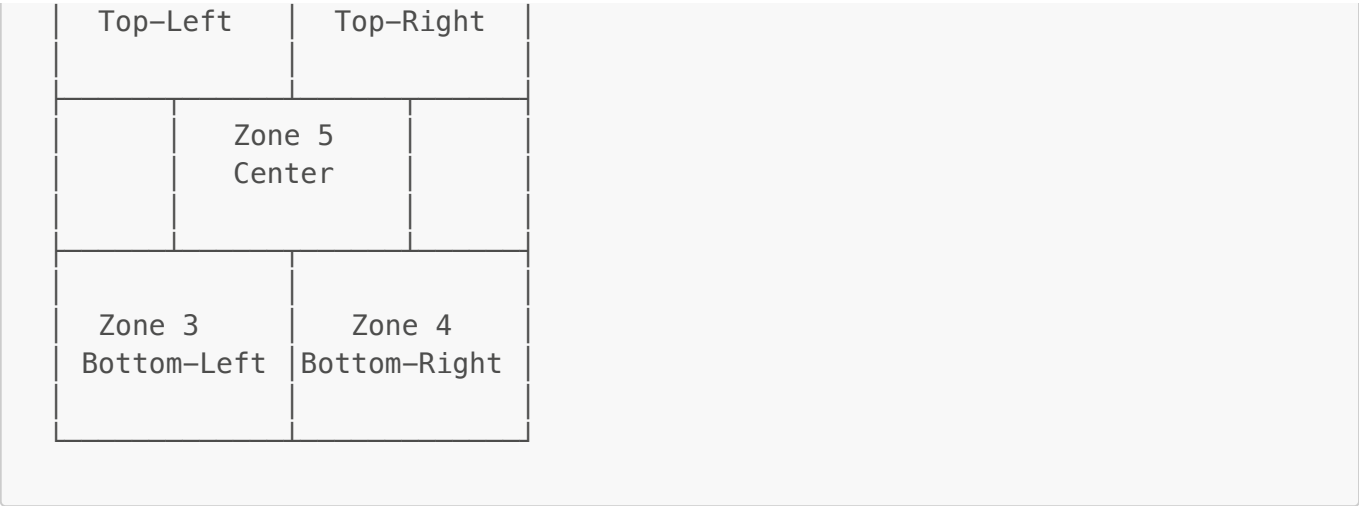
| Task                 | Description                                                                               |
|----------------------|-------------------------------------------------------------------------------------------|
| Image preprocessing  | CLAHE contrast enhancement + Gaussian noise reduction for dusty/uneven quarry lighting    |
| Stone segmentation   | Otsu thresholding + morphological operations to isolate individual stones from background |
| Watershed separation | Separate touching/overlapping stones into individual particles                            |
| Particle measurement | Calculate equivalent diameter of each stone in millimeters                                |
| Size distribution    | Compute min, max, mean, median, standard deviation across all detected particles          |
| Grade classification | Map particle size distribution to grade: 6mm, 10mm, 12mm, or 20mm                         |
| Confidence scoring   | Percentage-based confidence: how sure the system is about the classification              |

#### 1.2 Multi-Zone Analysis (5-Zone Sampling)

The system doesn't just look at one spot — it samples 5 zones across the truck bed to catch mixed loads or contamination.

Zone 1

Zone 2



| Task                    | Description                                                     |
|-------------------------|-----------------------------------------------------------------|
| Zone extraction         | Divide camera frame into 5 zones (4 quadrants + center)         |
| Per-zone classification | Each zone classified independently                              |
| Majority voting         | Final grade = majority vote across 5 zones (need 3+ agreement)  |
| Contamination detection | If zones disagree significantly → flag as "Mixed Load"          |
| Mismatch check          | Compare classified grade against expected grade (invoice/order) |

1.3 Camera Calibration System

| Task                         | Description                                                          |
|------------------------------|----------------------------------------------------------------------|
| Ruler detection              | Auto-detect ruler/scale bar in calibration image                     |
| Pixel-to-mm conversion       | Calculate exact px/mm ratio for the specific camera + distance setup |
| Calibration validation       | Bounds checking to reject bad calibrations                           |
| Manual calibration mode      | Operator can manually enter px/mm if auto-detection fails            |
| Per-gate calibration storage | Each gate/camera can have its own calibration profile                |

1.4 Testing with Actual Quarry Samples

| Task                          | Description                                                          |
|-------------------------------|----------------------------------------------------------------------|
| Collect sample images         | 50+ photos per grade from client's actual quarry (all 4 grades)      |
| Different lighting conditions | Morning, afternoon, overcast, shade, artificial light                |
| Wet vs dry material           | Stone looks different when wet — system must handle both             |
| Threshold tuning              | Adjust classification boundaries to match client's specific material |
| Accuracy validation           | Target: 90%+ correct classification across all grades                |
| Edge case handling            | Mixed loads, very dusty conditions, partial truck views              |

Phase 1 Deliverables

- ☐ Working classification engine for all 4 grades (6mm, 10mm, 12mm, 20mm)
- ☐ 5-zone analysis with contamination detection
- ☐ Camera calibration tool (auto + manual)
- ☐ Tested and tuned with client's actual material samples
- ☐ Accuracy report: grade-wise success rate

Milestone Demo

Show client: upload a photo of his quarry truck → system classifies all 5 zones, shows grade, confidence %, and whether it's mixed/contaminated.

Phase 2: Live Surveillance & Auto-Capture

**Duration:** Month 1–2  
**Investment:** ₹10,00,000  
**Depends on:** Phase 1

What This Phase Delivers

Transform the system from "upload a photo" to "cameras automatically capture and analyze every truck passing through the gate — 24/7, no human needed."

Hardware Required (Client's Cost, Separate)

| Item              | Specification                                               | Qty per Gate | Est. Cost      |
|-------------------|-------------------------------------------------------------|--------------|----------------|
| IP Camera         | Hikvision/Dahua, 2–5MP, PoE, IP66, IR night vision          | 1–2          | ₹8,000–15,000  |
| ANPR Camera       | Dedicated number plate camera (or second angle of main cam) | 1            | ₹10,000–20,000 |
| PoE Switch        | 4/8 port, powers cameras via LAN cable                      | 1            | ₹3,000–5,000   |
| On-site PC/Server | Intel i5+, 8GB+ RAM, 1TB HDD, Ubuntu/Windows                | 1            | ₹40,000–60,000 |
| UPS               | Protect PC from power cuts, 1–2 hour backup                 | 1            | ₹5,000–10,000  |
| LAN Cabling       | CAT6 cable from cameras to PC                               | As needed    | ₹2,000–5,000   |
| Mounting hardware | Camera brackets, poles, weatherproof housing                | As needed    | ₹3,000–5,000   |

**Total hardware per gate: ₹70,000–1,20,000**  
**For 3 gates: ₹2,10,000–3,60,000** (within the ₹3–5L estimate)

## Detailed Tasks

### 2.1 RTSP Camera Integration

| Task                     | Description                                                   |
|--------------------------|---------------------------------------------------------------|
| RTSP stream reader       | Connect to IP camera's live video feed via RTSP protocol      |
| Multi-camera support     | Handle 2–3 cameras simultaneously (one per gate)              |
| Frame extraction         | Pull individual frames from the live stream for analysis      |
| Camera health monitoring | Detect if camera goes offline, sends alert                    |
| Reconnection logic       | Auto-reconnect if camera feed drops (power cut, network blip) |
| Configuration per camera | Each camera has its own RTSP URL, resolution, angle settings  |

### 2.2 Automatic Truck Detection & Capture

| Task                        | Description                                                                  |
|-----------------------------|------------------------------------------------------------------------------|
| Motion detection            | Detect when a truck enters the camera's field of view                        |
| Truck presence confirmation | Distinguish truck from person/animal/vehicle passing by                      |
| Optimal frame capture       | Wait for truck to be fully in frame, then capture best frame                 |
| Multi-frame analysis        | Capture 3–5 frames per truck, pick the best quality one                      |
| Trigger logging             | Record exact timestamp of every capture event                                |
| Gate status tracking        | Know if gate is idle / truck approaching / truck in position / truck leaving |

### 2.3 24/7 Operation

| Task                 | Description                                            |
|----------------------|--------------------------------------------------------|
| Day mode             | Normal color image processing                          |
| Night mode (IR)      | Handle infrared/low-light images from camera's IR LEDs |
| Auto-switching       | System adapts processing based on lighting conditions  |
| Continuous operation | Run as a background service, auto-start on boot        |
| Crash recovery       | If software crashes, auto-restart within 30 seconds    |
| Log rotation         | Manage log files so disk doesn't fill up               |

### 2.4 On-Site Hardware Coordination

| Task                      | Description                                              |
|---------------------------|----------------------------------------------------------|
| Camera placement planning | Site visit — determine exact mounting positions per gate |

| Task               | Description                                            |
|--------------------|--------------------------------------------------------|
| Angle optimization | Camera angle to see truck bed clearly from 2–3m height |
| Network setup      | LAN cabling from cameras to on-site PC                 |
| PC setup           | Install OS, software, configure auto-start             |
| Testing            | Verify all cameras feeding correctly, day and night    |

## Phase 2 Deliverables

- ☐ Live RTSP feed working from all gate cameras
- ☐ Automatic truck detection and frame capture
- ☐ 24/7 operation with day/night mode
- ☐ Auto-restart on crash, reconnect on camera dropout
- ☐ On-site PC configured and all hardware installed
- ☐ Each gate captures + classifies automatically without human intervention

## Milestone Demo

Take client to the gate → truck passes → system auto-captures, classifies, and logs the result on screen. No button clicks, no human involvement. Show it working in both day and night.

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## Phase 3: Vehicle Identification (ANPR)

**Duration:** Month 2–3

**Investment:** ₹8,00,000

**Depends on:** Phase 2

## What This Phase Delivers

Every truck that passes is identified by its number plate — automatically. The system knows WHICH truck carried WHAT material, WHEN, and whether it was authorized.

## Detailed Tasks

### 3.1 Number Plate Recognition Engine

| Task                         | Description                                                            |
|------------------------------|------------------------------------------------------------------------|
| Plate detection              | Locate the number plate region in the camera frame                     |
| Plate extraction             | Crop and enhance the plate area for reading                            |
| OCR (character recognition)  | Read the plate text: e.g., "TN 38 AB 1234"                             |
| Indian plate format handling | Handle yellow commercial plates, white plates, different state formats |
| Dirty/damaged plate handling | Enhancement for dusty, mud-covered, bent plates (common in quarries)   |

| Task                     | Description                                               |
|--------------------------|-----------------------------------------------------------|
| Confidence scoring       | How sure the system is about the plate reading            |
| Manual correction option | If OCR fails, operator can manually type the plate number |

### 3.2 Vehicle Database

| Task               | Description                                                                |
|--------------------|----------------------------------------------------------------------------|
| Vehicle registry   | Store all known trucks: plate number, owner, capacity, authorized material |
| Driver database    | Link drivers to vehicles (optional — can add later)                        |
| Authorization list | Which trucks are authorized to pick up which material                      |
| Blacklist          | Flag specific vehicles (known offenders, banned trucks)                    |
| Auto-learn         | New plates automatically added to database on first appearance             |
| Search & filter    | Look up any vehicle by plate number, date range, material carried          |

### 3.3 Entry/Exit Matching

| Task                     | Description                                                     |
|--------------------------|-----------------------------------------------------------------|
| Entry detection          | Log truck entering the quarry (empty)                           |
| Exit detection           | Log truck leaving the quarry (loaded)                           |
| Entry-exit pairing       | Match each exit with its corresponding entry                    |
| Unauthorized exit alert  | Truck exits loaded but never entered properly → immediate alert |
| Time tracking            | How long each truck spent inside the quarry                     |
| Multiple trips detection | Flag trucks making unusually many trips per day                 |

### 3.4 Repeat Offender Tracking

| Task                   | Description                                                       |
|------------------------|-------------------------------------------------------------------|
| Anomaly scoring        | Each vehicle gets a risk score based on history                   |
| Flagging rules         | Auto-flag: too many trips, odd timing (late night), mismatches    |
| Weekly offender report | Top 10 suspicious vehicles of the week                            |
| Pattern detection      | Same truck, same time, same gate — habitual unauthorized dispatch |

### Phase 3 Deliverables

- ☐ Automatic number plate reading for every passing truck
- ☐ Vehicle database with authorization management
- ☐ Entry/exit matching with unauthorized exit alerts

- ☐ Repeat offender tracking with risk scoring
- ☐ Searchable vehicle history log

Milestone Demo

Truck passes the gate → screen shows: plate "TN 38 AB 1234", material "12mm", authorized "YES", confidence 94%. Show the vehicle history for that truck — all past visits, materials carried, timestamps.

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## Phase 4: Material Reconciliation & Theft Detection

**Duration:** Month 3–4  
**Investment:** ₹8,00,000  
**Depends on:** Phase 2, Phase 3

What This Phase Delivers

The "catch the thief" engine. Cross-references all data — production, dispatch, weighbridge, truck logs — to find exactly where material is going missing, how much, and when.

Detailed Tasks

4.1 Weighbridge Data Integration

| Task                       | Description                                                             |
|----------------------------|-------------------------------------------------------------------------|
| Weighbridge data import    | Read data from existing weighbridge system (CSV/Excel/database)         |
| Weight per truck           | Gross weight, tare weight, net material weight                          |
| Material weight estimation | Cross-reference: camera says "12mm" + weighbridge says "18 tonnes"      |
| Weight anomaly detection   | Expected weight for 12mm truck = 15–20T. Below 12T or above 22T → alert |
| Auto-matching              | Match weighbridge entry to camera capture by truck plate + timestamp    |
| Manual weighbridge entry   | If no digital weighbridge, operator can type weight manually            |

4.2 Daily Material Balance

| Task              | Description                                                                    |
|-------------------|--------------------------------------------------------------------------------|
| Production input  | Daily production numbers per grade (entered by manager or imported)            |
| Dispatch tracking | Auto-calculated from camera + weighbridge data                                 |
| Stock calculation | Stock = Yesterday's stock + Today's production - Today's dispatch              |
| Balance report    | One-page view: produced, dispatched, stock remaining — per grade               |
| Gap detection     | If dispatch > production + stock → material coming from somewhere unauthorized |
| Historical trend  | Weekly/monthly balance trend — is the gap growing or shrinking?                |

### 4.3 Theft Alert Engine

| Task                        | Description                                                                              |
|-----------------------------|------------------------------------------------------------------------------------------|
| Real-time anomaly detection | Triggered whenever data doesn't add up                                                   |
| Alert types:                |                                                                                          |
| — Unauthorized dispatch     | Truck left with material but no corresponding order                                      |
| — Grade mismatch            | Order says 12mm but camera detects 20mm loaded                                           |
| — Weight anomaly            | Load weight doesn't match expected for that grade                                        |
| — After-hours dispatch      | Truck left between 10 PM – 6 AM (configurable)                                           |
| — Excess trips              | More trucks dispatched than orders placed                                                |
| — Stock discrepancy         | Stock count doesn't match calculated stock                                               |
| Alert severity              | Low (info) / Medium (review needed) / High (likely theft) / Critical (confirmed anomaly) |
| Alert delivery              | On-screen notification + SMS/WhatsApp to owner                                           |
| False positive handling     | Owner can mark alerts as reviewed/false alarm — system learns patterns                   |

### 4.4 Shift-Wise Accountability

| Task                      | Description                                                              |
|---------------------------|--------------------------------------------------------------------------|
| Shift definition          | Morning (6AM–2PM), Afternoon (2PM–10PM), Night (10PM–6AM) — configurable |
| Per-shift dispatch report | How much material left during each shift                                 |
| Shift comparison          | Compare shifts against each other — which shift has most anomalies?      |
| Gate operator log         | Who was on duty when an anomaly occurred                                 |
| Shift handover report     | Summary at each shift change — totals, alerts, pending actions           |

### Phase 4 Deliverables

- ☐ Weighbridge data integration (auto-import or manual entry)
- ☐ Daily material balance: production vs dispatch vs stock
- ☐ Real-time theft alert engine with 6+ anomaly types
- ☐ Alert severity levels with SMS/WhatsApp notification
- ☐ Shift-wise accountability reports
- ☐ Historical trend analysis

### Milestone Demo

Show the owner: "Today, 45 trucks dispatched, 680 tonnes total. Production was 650 tonnes. Stock should be 200 tonnes but calculated stock is 170 tonnes. 30 tonnes gap — here are the 3 suspicious dispatches flagged with HIGH severity."

## Phase 5: Owner Dashboard, Mobile Access & Data Security

**Duration:** Month 3–4 (parallel with Phase 3–4)  
**Investment:** ₹7,00,000  
**Depends on:** Phase 1 (builds progressively as other phases complete)

### What This Phase Delivers

The control center — everything the owner needs to monitor his quarry from anywhere, plus the encrypted data security architecture that protects all intelligence data.

### Detailed Tasks

#### 5.1 Owner Dashboard (Web)

| Task                | Description                                                                  |
|---------------------|------------------------------------------------------------------------------|
| Real-time overview  | Live status of all gates, current truck count, today's dispatch total        |
| Gate-wise view      | Click on any gate → see live feed, last 10 trucks, current status            |
| Material breakdown  | Pie chart / bar chart: how much of each grade dispatched today               |
| Truck log           | Scrollable table: timestamp, plate, grade, weight, gate, status (pass/alert) |
| Alert panel         | Active alerts with severity badges, one-click review/dismiss                 |
| Reconciliation view | Production vs dispatch vs stock — with gap highlighted in red                |
| Date range filter   | View any historical period — today, this week, this month, custom range      |
| Export to PDF       | One-click report download for any date range                                 |

#### 5.2 Mobile-Optimized View

| Task               | Description                                                               |
|--------------------|---------------------------------------------------------------------------|
| Responsive design  | Dashboard works on phone browser — no app install needed                  |
| Key metrics on top | Today's dispatch count, total tonnes, active alerts — visible at a glance |
| Push-style alerts  | SMS or WhatsApp message with alert details + link to dashboard            |
| Quick actions      | Approve/dismiss alerts from phone                                         |
| Offline summary    | Daily summary SMS even if owner doesn't open the dashboard                |

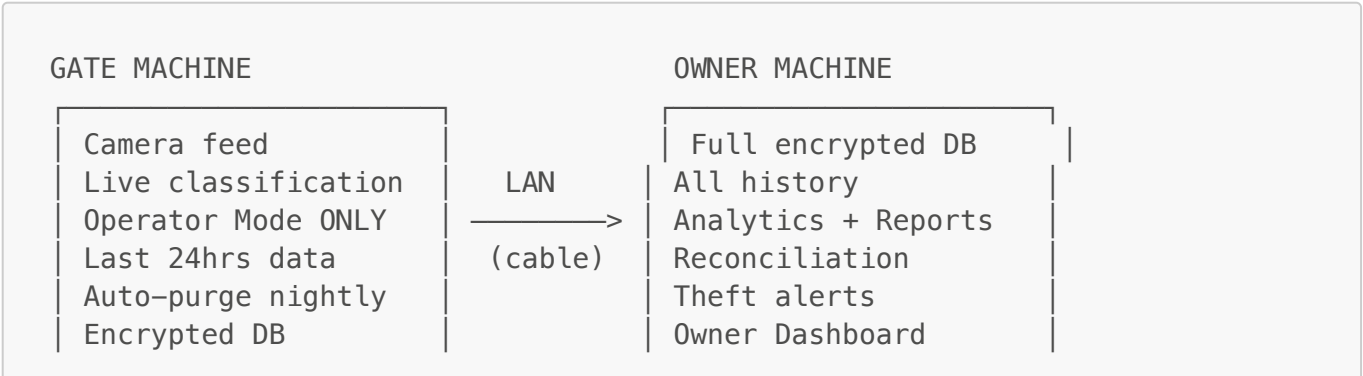
#### 5.3 Role-Based Access

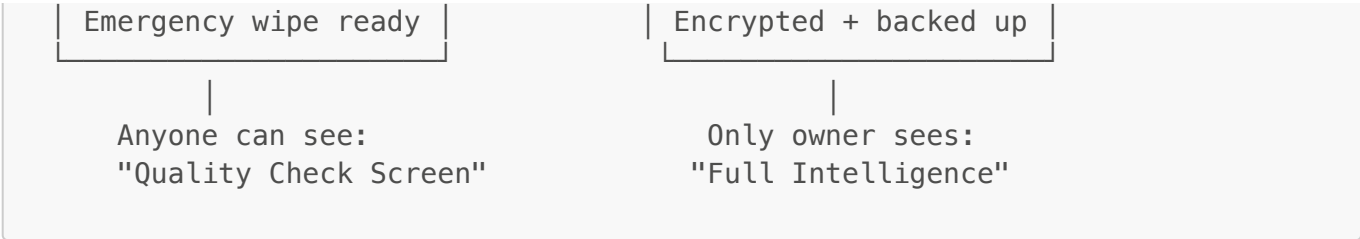
| Role          | Can See                                                                 | Can Do                                                           |
|---------------|-------------------------------------------------------------------------|------------------------------------------------------------------|
| Owner         | Everything — full analytics, reconciliation, theft reports, all history | Configure system, manage users, download reports, dismiss alerts |
| Manager       | Today's dispatch, truck log, basic alerts                               | Acknowledge alerts, add manual entries                           |
| Gate Operator | Current truck status, pass/fail result only                             | Nothing — view only, cannot modify any data                      |

| Task               | Description                                      |
|--------------------|--------------------------------------------------|
| Login system       | Username + password per user                     |
| Role assignment    | Owner assigns roles to manager/operator accounts |
| Audit trail        | Log who accessed what and when                   |
| Session management | Auto-logout after inactivity                     |

5.4 Data Security Architecture

| Task                | Description                                                                     |
|---------------------|---------------------------------------------------------------------------------|
| Encrypted database  | SQLCipher encryption — database file is unreadable without password             |
| Split storage       | Gate machine keeps 24hrs only; Owner machine keeps full history                 |
| Auto-purge          | Gate machine deletes records older than 24–48 hours automatically               |
| LAN sync            | Data transfers from gate machine to owner machine via local cable — no internet |
| Dual dashboard mode | Gate shows Operator Mode (basic); Owner machine shows full analytics            |
| Hidden owner access | No visible button for owner mode on gate machine — accessed only via PIN        |
| Auto-lock           | System locks after 5 minutes of inactivity                                      |
| Emergency wipe      | Hidden keyboard shortcut to instantly clear gate machine data                   |
| Tamper-proof logs   | Append-only log file — entries cannot be deleted or modified                    |
| Backup system       | Automatic encrypted backup to USB drive (optional secondary backup)             |





5.5 Alert & Notification System

| Task                    | Description                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------|
| SMS integration         | Twilio or local SMS gateway for alerts                                                          |
| WhatsApp integration    | WhatsApp Business API for rich alerts with images                                               |
| Alert templates         | Pre-formatted messages: "🚚 Unauthorized dispatch at Gate 2 — TN38AB1234 — 12mm — 18T — 2:34 AM" |
| Alert grouping          | Don't spam — group similar alerts within 15 minutes                                             |
| Daily digest            | End-of-day summary: total trucks, tonnes, alerts, gaps                                          |
| Configurable recipients | Owner + optional manager phone numbers                                                          |

Phase 5 Deliverables

- ☐ Full web dashboard with real-time monitoring
- ☐ Mobile-optimized view accessible from phone browser
- ☐ Role-based access (Owner / Manager / Operator)
- ☐ Encrypted database (SQLCIPHER) on both machines
- ☐ Split storage with auto-purge on gate machine
- ☐ LAN sync between gate and owner machine
- ☐ Dual dashboard mode (Operator / Owner)
- ☐ Emergency wipe capability
- ☐ SMS/WhatsApp alert system
- ☐ PDF report generation and export

Milestone Demo

Give owner his phone → show him the dashboard with today's data. Trigger a test alert → WhatsApp pops up on his phone within seconds. Open the gate machine → show him it only has today's basic data. Open the owner machine → show him full history, analytics, reconciliation. Explain: "This is what anyone sees at the gate. This is what only you see."

Phase 6: Deployment, Training & Support

**Duration:** Month 4–5  
**Investment:** ₹5,00,000

**Depends on:** All previous phases

What This Phase Delivers

The system goes live at the quarry. Hardware installed, software configured, operators trained, owner confident. Plus 6 months of support after go-live.

Detailed Tasks

6.1 Site Preparation & Hardware Installation

| Task             | Description                                                            |
|------------------|------------------------------------------------------------------------|
| Site survey      | Visit each gate — finalize camera positions, cable routes, PC location |
| Camera mounting  | Install cameras at 2–3m height at each gate, weatherproof housing      |
| Cable routing    | CAT6 LAN cables from cameras to gate PC, gate PC to owner PC           |
| PC setup         | Install OS, Q-Vision software, configure auto-start on boot            |
| UPS installation | Connect PC and cameras to UPS for power-cut protection                 |
| Network testing  | Verify all camera feeds stable, LAN sync working                       |
| Night testing    | Verify IR cameras working after dark                                   |

6.2 System Configuration

| Task                      | Description                                                              |
|---------------------------|--------------------------------------------------------------------------|
| Per-gate calibration      | Run calibration tool at each gate with ruler                             |
| Grade threshold tuning    | Fine-tune classification thresholds with client's actual material        |
| Alert rules configuration | Set working hours, shift times, weight thresholds per client preferences |
| User accounts             | Create owner, manager, operator accounts with correct roles              |
| Backup schedule           | Configure automatic encrypted backup (USB or second machine)             |
| Auto-purge schedule       | Set 24hr or 48hr purge window on gate machines                           |

6.3 Operator Training

| Session   | Audience       | Duration | Content                                                                                         |
|-----------|----------------|----------|-------------------------------------------------------------------------------------------------|
| Session 1 | Gate operators | 2 hours  | How the gate screen works, what pass/fail means, when to call manager                           |
| Session 2 | Manager(s)     | 2 hours  | Dashboard overview, alert handling, daily report review, manual entry                           |
| Session 3 | Owner          | 1 hour   | Full system walkthrough, owner dashboard, mobile access, data security features, emergency wipe |

| Training Material             | Description                                                                      |
|-------------------------------|----------------------------------------------------------------------------------|
| Operator quick-reference card | Laminated A4 sheet at each gate — common scenarios and what to do                |
| Manager guide                 | 5-page PDF — dashboard navigation, alert handling, reporting                     |
| Owner guide                   | 3-page PDF — mobile access, key metrics, how to check data, emergency procedures |
| Troubleshooting guide         | Common issues: camera offline, system frozen, power cut recovery                 |

#### 6.4 Go-Live & Burn-In Period

| Task                  | Description                                                             |
|-----------------------|-------------------------------------------------------------------------|
| Soft launch (Week 1)  | System runs alongside existing manual process — both capture data       |
| Comparison validation | Compare system results vs manual records — verify accuracy              |
| Issue fixing          | Address any bugs or edge cases found during soft launch                 |
| Full launch (Week 2)  | System takes over as primary monitoring — manual process becomes backup |
| 2-week monitoring     | Developer monitors remotely, daily check-in with manager                |

#### 6.5 Post-Deployment Support (6 Months)

| Support Type                            | Response Time   | Includes                             |
|-----------------------------------------|-----------------|--------------------------------------|
| Critical (system down)                  | Within 2 hours  | Remote fix or on-site visit          |
| High (alerts not working, camera issue) | Within 8 hours  | Remote diagnosis + fix               |
| Medium (report incorrect, minor bug)    | Within 24 hours | Remote fix                           |
| Low (feature request, UI change)        | Within 1 week   | Evaluation + implementation if minor |

##### Support Includes

|                      |                                                        |
|----------------------|--------------------------------------------------------|
| Bug fixes            | Any defects found post-deployment                      |
| Performance tuning   | If accuracy drops due to seasonal changes (rain, dust) |
| Camera recalibration | If cameras are moved or replaced                       |
| Software updates     | Minor improvements and stability fixes                 |
| Remote monitoring    | Periodic check-in to ensure system health              |
| Phone support        | WhatsApp/call support for manager and owner            |

##### Support Does NOT Include

|                         |                                                |
|-------------------------|------------------------------------------------|
| New feature development | Quoted separately                              |
| Hardware replacement    | Client's responsibility (cameras, server, UPS) |

| Support Does NOT Include |                                            |
|--------------------------|--------------------------------------------|
| Internet-related issues  | System is offline — no internet dependency |
| Operator errors          | Covered by re-training, not software fix   |

Phase 6 Deliverables

- ☐ All cameras installed and tested at every gate
- ☐ On-site PC configured with auto-start and crash recovery
- ☐ System calibrated for each gate with client's material
- ☐ All user accounts created with correct roles
- ☐ 3 training sessions completed (operators, manager, owner)
- ☐ Training materials delivered (printed + PDF)
- ☐ 2-week soft launch completed with validation report
- ☐ Full system go-live
- ☐ 6-month support agreement active

Milestone

Owner receives a WhatsApp message: "Q-Vision is LIVE. Gate 1:  Online. Gate 2:  Online. Gate 3:  Online. Today's first truck: TN38AB1234, 12mm, 18.5T, 6:42 AM. All systems normal."

Complete Deliverables Summary

| #  | Deliverable                                        | Phase   |
|----|----------------------------------------------------|---------|
| 1  | Stone grade classification (6mm, 10mm, 12mm, 20mm) | Phase 1 |
| 2  | 5-zone analysis with contamination detection       | Phase 1 |
| 3  | Camera calibration system (auto + manual)          | Phase 1 |
| 4  | Accuracy validation report with client's material  | Phase 1 |
| 5  | Live RTSP camera integration (multi-gate)          | Phase 2 |
| 6  | Automatic truck detection and frame capture        | Phase 2 |
| 7  | 24/7 operation with day/night mode                 | Phase 2 |
| 8  | Auto-restart and crash recovery                    | Phase 2 |
| 9  | Number plate recognition (ANPR)                    | Phase 3 |
| 10 | Vehicle database with authorization management     | Phase 3 |
| 11 | Entry/exit matching with unauthorized alerts       | Phase 3 |
| 12 | Repeat offender tracking                           | Phase 3 |
| 13 | Weighbridge data integration                       | Phase 4 |

| #  | Deliverable                                              | Phase   |
|----|----------------------------------------------------------|---------|
| 14 | Daily material balance (produced vs dispatched vs stock) | Phase 4 |
| 15 | Theft alert engine (6+ anomaly types)                    | Phase 4 |
| 16 | Shift-wise accountability reports                        | Phase 4 |
| 17 | Web dashboard with real-time monitoring                  | Phase 5 |
| 18 | Mobile-optimized view                                    | Phase 5 |
| 19 | Role-based access (Owner / Manager / Operator)           | Phase 5 |
| 20 | Encrypted database (SQLCipher)                           | Phase 5 |
| 21 | Split storage with auto-purge                            | Phase 5 |
| 22 | Dual dashboard mode (Operator / Owner)                   | Phase 5 |
| 23 | SMS/WhatsApp alert system                                | Phase 5 |
| 24 | PDF report generation                                    | Phase 5 |
| 25 | Emergency data wipe                                      | Phase 5 |
| 26 | On-site hardware installation                            | Phase 6 |
| 27 | Per-gate calibration and tuning                          | Phase 6 |
| 28 | Operator, Manager, Owner training                        | Phase 6 |
| 29 | Soft launch + validation                                 | Phase 6 |
| 30 | 6-month post-deployment support                          | Phase 6 |

## Risk Register

| Risk                               | Impact                          | Mitigation                                                      |
|------------------------------------|---------------------------------|-----------------------------------------------------------------|
| Camera gives poor image quality    | Classification accuracy drops   | Test cameras before bulk purchase; use 4MP+ resolution          |
| Dusty/muddy number plates          | ANPR fails to read              | Image enhancement pipeline; manual correction fallback          |
| Power cuts at quarry               | System goes down                | UPS backup (1–2 hrs); auto-start on power resume                |
| Night lighting poor                | Night classification unreliable | IR cameras with built-in LEDs; test night mode during Phase 2   |
| Client's material varies by season | Thresholds need re-tuning       | Phase 6 support includes recalibration; configurable thresholds |
| Operator resistance                | Staff may try to bypass system  | Training + owner oversight + tamper-proof logs                  |

| Risk                 | Impact                    | Mitigation                                           |
|----------------------|---------------------------|------------------------------------------------------|
| Network cable damage | Gate-to-owner sync breaks | Use protected cable conduit; USB backup as secondary |

Investment Summary

| Phase                                      | Amount                | Payment Trigger          |
|--------------------------------------------|-----------------------|--------------------------|
| Phase 1: Core Intelligence Engine          | ₹12,00,000            | Included in 40% advance  |
| Phase 2: Live Surveillance & Auto-Capture  | ₹10,00,000            | Phase 1–2 demo delivery  |
| Phase 3: Vehicle Identification (ANPR)     | ₹8,00,000             | Phase 3–5 delivery       |
| Phase 4: Reconciliation & Theft Detection  | ₹8,00,000             | Phase 3–5 delivery       |
| Phase 5: Dashboard, Mobile & Data Security | ₹7,00,000             | Phase 3–5 delivery       |
| Phase 6: Deployment, Training & Support    | ₹5,00,000             | Final deployment         |
| <b>Total</b>                               | <b>₹50,00,000</b>     |                          |
| Hardware (client's cost, separate)         | ₹3,00,000 – ₹5,00,000 | Client procures directly |

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